2 Sample Location Problem

Aniv Mazumder (MD2505)
Koushan Saha (MD2513)
Sanjhali Parmar (MD2521)

Indian Statistical Institute
M.Stat, 2nd Semester

March 2026

Location Shift Model and Hypotheses

Location Shift Model

Let $X_1, \ldots, X_m \overset{iid}{\sim} F(x)$ and $Y_1, \ldots, Y_n \overset{iid}{\sim} G(x)$, be two independent samples where F is continuous.
The shift model is defined as: $G(x) = F(x - \theta)$

Hypothesis Setups

1. Location Shift direction

- $H_0 : \theta = 0$
- $H_1 : \theta > 0$
- $H_2 : \theta < 0$
- $H_3 : \theta \neq 0$

2. Distributional

- $H_0 : F = G$
- $H_1 : G \leq F$
- $H_2 : G \geq F$
- $H_3 : F \neq G$

3. Stochastic

- $H_0 : X \overset{d}{=} Y$
- $H_1 : X <_{st} Y$
- $H_2 : Y <_{st} X$
- $H_3 : X \overset{d}{\neq} Y$

Test Statistic

The **Mann-Whitney U statistic**, defined as

$$U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(X_i > Y_j)$$

Decision Rule: Under $H_1 : \theta > 0$, we expect Y to be generally larger than X. Therefore, we **reject H_0 for small values of U**.

Relationship with Other Rank Statistics

Rejection Criteria (at level α)

Using the Mann-Whitney U statistic:

Alternative	**Rejection Region for H_0**
$H_1 : \theta > 0$	Reject H_0 if $U \leq c_\alpha$ (Small values)
$H_2 : \theta < 0$	Reject H_0 if $U \geq c'_\alpha$ (Large values)
$H_3 : \theta \neq 0$	Reject H_0 if $U \leq c_{\alpha_1}$ or $U \geq c'_{\alpha_2}$

Wilcoxon Rank Sum Statistic (W)

W is the sum of ranks of the X sample in the pooled data. The two are linearly related:

$$W = U + \frac{m(m+1)}{2}$$

Two-Sample Kruskal-Wallis Stat (H)

For $k = 2$ samples, the Kruskal-Wallis H statistic is a quadratic function of the Mann-Whitney U (centered at its mean):

$$H = \frac{12}{N(N+1)mn}\left(U - \frac{mn}{2}\right)^2 = \frac{(U - E(U))^2}{Var(U)} \cdot \frac{m+n+1}{m+n}$$

Conclusion: These statistics provide identical p-values and lead to the same statistical decisions.

Statistical Properties under H_0

Moments of the U Statistic

Under H_0, the Mann-Whitney U statistic has:

- **Mean:** $E(U) = \frac{mn}{2}$
- **Variance:** $Var(U) = \frac{mn(m+n+1)}{12}$

Distribution-Free Property

Under H_0, since all $N!$ permutations of ranks are equally likely, the distribution of U depends only on the sample sizes m and n, making it **distribution-free**.

Asymptotic Normality

As $N = m + n \to \infty$, the standardized statistic converges in distribution:

$$Z_N = \frac{U - E(U)}{\sqrt{Var(U)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Real-life Example: Sputum Histamine Levels

Problem Statement and Data

We investigate whether sputum histamine levels ($\mu g/g$) differ between allergic and nonallergic individuals.

Group	Observations
X: Allergics ($m = 9$)	67.6, 39.6, 165.0, 100.0, 65.9, 111.2, 31.0, 102.5, 64.7
Y: Nonallergics ($n = 10$)	57.6, 36.6, 65.0, 130.0, 45.9, 101.8, 51.0, 92.5, 54.7, 85.1

Hypothesis Formulation

Let θ be the location shift parameter such that $F_Y(x) = F_X(x - \theta)$. To test if allergics have higher histamine levels:

- **Null Hypothesis (H_0):** $\theta = 0$ (No difference in distributions)
- **Alternative Hypothesis (H_1):** $\theta < 0$ (Allergics have higher levels)

Decision Rule

We evaluate this using Mann-Whitney U statistic. The criterion is to reject H_0 for sufficiently large values of U. Thus, the rejection region is of the form $\{U \geq c_\alpha\}$, where the critical value c_α is chosen to control the Type I error rate at the nominal level α.

Exact Distribution under H_0

Exact Distribution for $m = 7, n = 5$

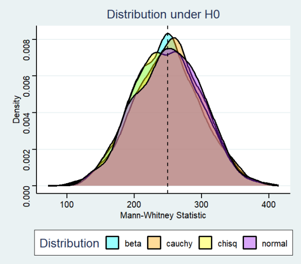
Under H_0, all rank combinations are equally likely. For $N = 12$:

- **Total Combinations:** $\binom{N}{n} = \binom{12}{5} = 792$
- **Range of U:** $[0, mn] = [0, 35]$ with symmetry about 17.5.

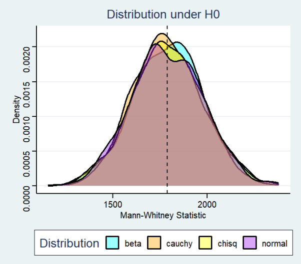
Probability Mass Function $P(U = u)$ (Under H_0)

u	$P(U = u)$	u	$P(U = u)$	u	$P(U = u)$	u	$P(U = u)$
0	0.00126	9	0.02652	18	0.06187	27	0.02146
1	0.00126	10	0.03283	19	0.06061	28	0.01641
2	0.00253	11	0.03788	20	0.05808	29	0.01263
3	0.00379	12	0.04419	21	0.05429	30	0.00884
4	0.00631	13	0.04924	22	0.04924	31	0.00631
5	0.00884	14	0.05429	23	0.04419	32	0.00379
6	0.01263	15	0.05808	24	0.03788	33	0.00253
7	0.01641	16	0.06061	25	0.03283	34	0.00126
8	0.02146	17	0.06187	26	0.02652	35	0.00126

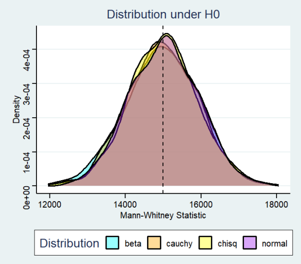
Simulation: Distribution-Free Property under H_0 (1000 iterations)



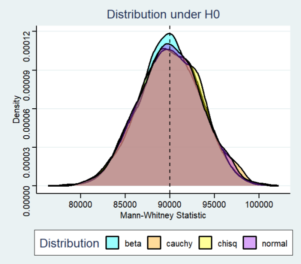
$m = 10$, $n = 50$



$m = 65$, $n = 55$



$m = 200$, $n = 150$



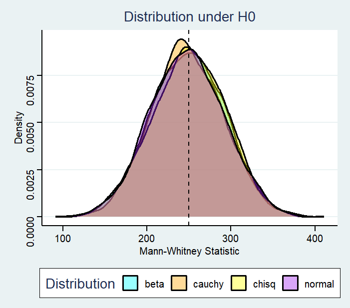
$m = 400$, $n = 450$

Key Interpretations of Simulations

Simulation Setup

1000 iterations drawn from Beta($\frac{1}{2}$, $\frac{1}{2}$), Cauchy(0,1), Normal(0,9), and Chi-square(4). The dashed vertical line represents:

$$E(U) = \frac{mn}{2}$$

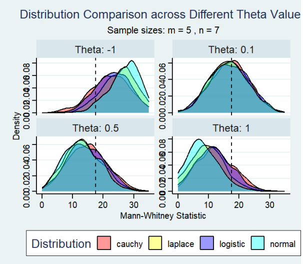


$m = 20$, $n = 25$

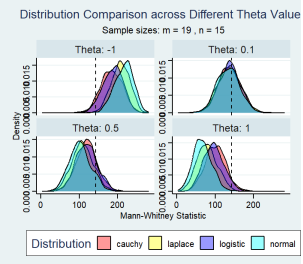
Observations & Conclusion

- **Centering:** In all cases, the density curves correctly center around the theoretical mean $\frac{mn}{2}$.
- **Distribution-Free:** The curves from entirely different underlying distributions (Beta, Cauchy, Normal, χ^2) almost perfectly overlap. The sampling distribution under H_0 is independent of the parent population.
- **Sample Size Effect:** As m and n increase, the overlap becomes more pronounced and the curves become significantly smoother.
- **Asymptotic Normality:** For larger sample sizes, the distributions visually converge to a symmetric bell shape, supporting the theoretical asymptotic normality.

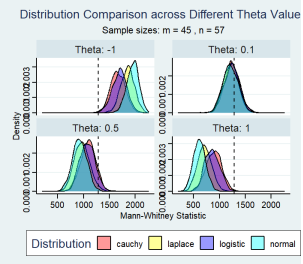
Simulations under H_1: Location Shift Models



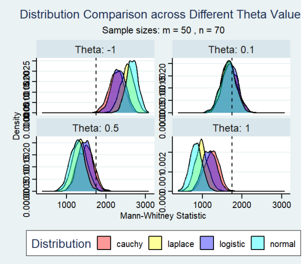
$(m, n) = (5, 7)$



$(m, n) = (19, 15)$



$(m, n) = (45, 57)$



$(m, n) = (50, 70)$

Key Interpretations of Simulations under H_1

Effect of Shift (θ) and Sample Size

- **Power Scaling:** Displacement from the null expectation ($mn/2$) increases with both shift magnitude ($|\theta|$) and sample size, directly increasing test power.
- **Small vs. Large Samples:** At very small sizes $(5, 7)$, densities overlap heavily due to low resolving power. Large samples $(50, 70)$ show sharp separation and high power at $|\theta| = 1$.
- **Symmetry:** The U statistic shifts symmetrically above and below $mn/2$ for positive and negative shifts ($+\theta$ and $-\theta$), respectively.

Impact of Distributional Tail Weight

- **Light Tails (Normal, Laplace):** Yield highly concentrated densities and the strongest displacement, implying the highest power.
- **Near-Null Robustness:** At a minimal shift ($\theta = 0.1$), densities from all four families heavily overlap near $mn/2$, demonstrating that near-null behavior remains largely distribution-free.

Under the Null Hypothesis (H_0)

Let $X_1, \ldots, X_m \overset{iid}{\sim} F$ and $Y_1, \ldots, Y_n \overset{iid}{\sim} F$ be two independent samples. As $m, n \to \infty$ with $m/N \to \lambda \in (0, 1)$, the standardized statistic converges in distribution to a standard normal:

$$Z = \frac{U - \frac{mn}{2}}{\sqrt{\frac{mn(N+1)}{12}}} \xrightarrow{d} N(0, 1)$$

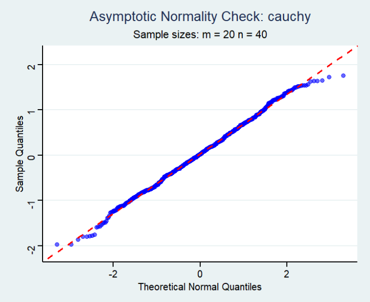
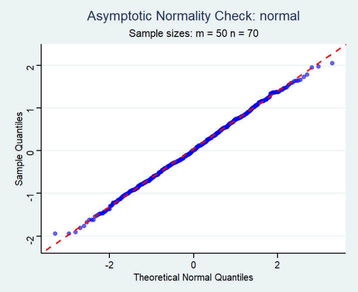
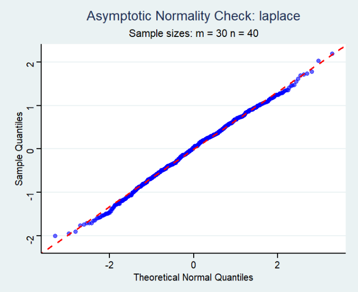
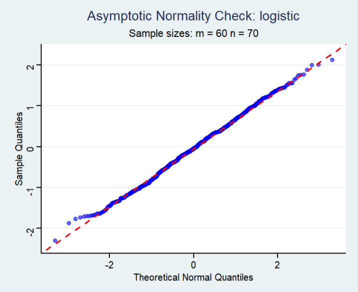


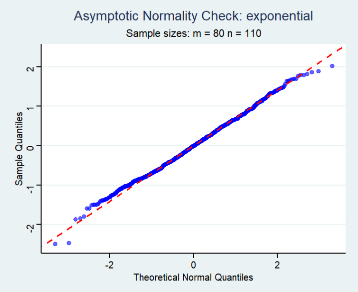
Figure: Normal Q-Q Plot 1

Q-Q Plots: Assessing Normality









Histograms and Q-Q Observations under H_0

Q-Q Plot Observations: The sample quantiles closely follow the theoretical normal diagonal line, indicating strong agreement with the normal distribution as sample sizes increase.

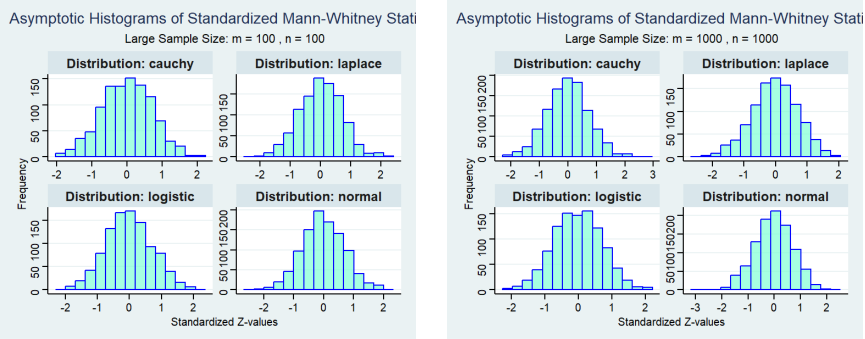


Figure: Standardized U Statistic under H_0

Observations of Histograms under H_0

The empirical distribution of the standardized U statistic is perfectly symmetric, bell-shaped, and visually matches the overlaid standard normal probability density curve.

Histograms and Observations under H_1 $(\theta > 0)$

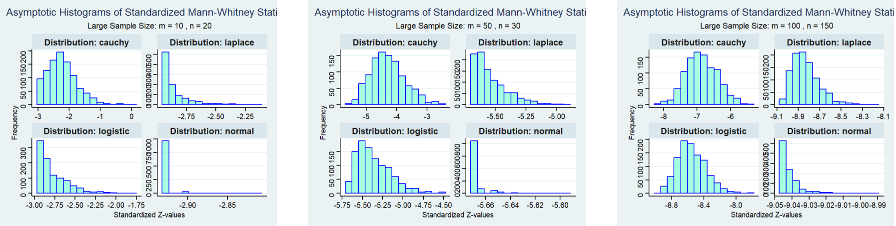
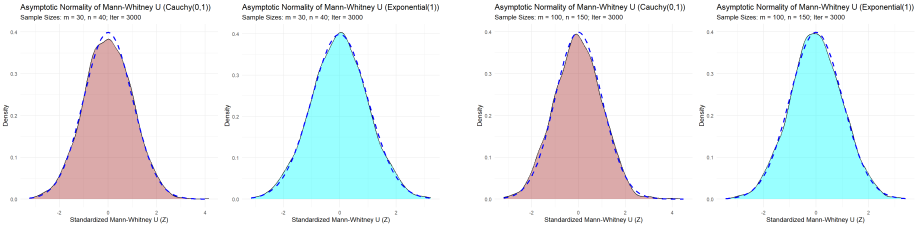


Figure: Distribution of U under Shift Alternatives

Observations under H_1: The Boundary Effect

Contrary to the null case, under a fixed and sufficiently large location shift $\theta = 5$, the asymptotic normality of U breaks down. Because U is bounded ($0 \leq U \leq mn$), strong sample separation pushes the statistic toward its limits. The histograms reveal severe right-skewness (particularly for lighter-tailed distributions like Normal and Logistic), illustrating that when $P(X < Y)$ approaches 0 or 1, the distribution becomes degenerate and highly non-normal. The Cauchy distribution's heavy tails cause persistent sample overlap, preventing extreme probabilities and maintaining histogram symmetry despite large location shifts.

Asymptotic Normal Approximation Overlays



These density overlays map the convergence speed, confirming that the continuous normal approximation smooths out the discrete steps of the underlying rank-sum distribution.

Interpretation of Asymptotic Overlays

- **Convergence Dynamics:** The density overlays illustrate that as sample sizes increase from ($m = 30$, $n = 40$) to ($m = 100$, $n = 150$), the empirical distribution of the standardized U statistic (Z) aligns more perfectly with the theoretical $N(0, 1)$ curve.
- **Distributional Invariance:** The normal approximation remains highly effective regardless of the parent distribution's shape—validating the test for both heavy-tailed (Cauchy) and skewed (Exponential) data under H_0.

Level Maintenance of the Mann-Whitney Test

Theoretical Properties

- **Distribution-Free:** Under $H_0 : F = G$ for continuous F, the null distribution of U depends only on m and n, guaranteeing exact level control.
- **Discrete Distributions:** In the presence of ties (e.g., Poisson), rankings are no longer equally likely. The test becomes **conservative**.

Simulation Steps

- **Step 1:** Generate $N_{\text{sim}} = 10,000$ iterations by drawing samples of size m and n from the same parent distribution (H_0).
- **Step 2 & 3:** Calculate the p-value for each iteration and count those less than the assigned level α: $\sum_{i=1}^{N_{\text{sim}}} \mathbf{1}(p_i < \alpha)$.
- **Step 4:** Divide this count by 10,000 to find the empirical size estimator: $\widehat{\alpha} = \frac{1}{N_{\text{sim}}} \times \text{count}$.
- **Step 5:** Check if $\widehat{\alpha}$ falls within the tolerance interval $\alpha \pm (1.96 \times \text{SE})$ to verify if the level is maintained.

Simulation Setup

- **Parameters:** $m = 11$, $n = 13$; $N_{\text{sim}} = 10,000$; $\alpha \in \{0.05, 0.01\}$.
- **Tested:** Normal(0,1), Cauchy(0,1), Poisson($\lambda = 10$).

Table: Estimated Size $\widehat{\alpha}$ under H_0

Distribution	Nominal α	Estimated $\widehat{\alpha}$	Tolerance?
Normal(0,1)	0.05	0.0486	✓
Normal(0,1)	0.01	0.0088	✓
Cauchy(0,1)	0.05	0.0451	✓
Cauchy(0,1)	0.01	0.0094	✓
Poisson(10)	0.05	0.0457	✓
Poisson(10)	0.01	0.0087	✓

Key Observations

- **Exact Maintenance:** Continuous distributions fall within the 95% CI $[0.0457, 0.0543]$ for $\alpha = 0.05$.
- **Conservative Behavior:** The Poisson result ($\widehat{\alpha} \approx 0.0457$) reflects ties from discreteness.
- **Robustness:** Maintenance holds regardless of tail behavior or skewness.

The Hodges-Lehmann Estimator (HLE) and CI of θ

Definition and Asymptotic Properties

The HLE $\hat{\theta}$ is the point estimator of the location shift θ associated with the Wilcoxon Rank-Sum test. It is defined as the **median of all possible pairwise differences**:

$$\hat{\theta} = \text{median}\{D_{i,j} = Y_j - X_i : 1 \leq i \leq m, 1 \leq j \leq n\}$$

Confidence Interval and Simulation Setup for Performance Evaluation

- **Confidence Interval:** A $(1-\alpha)$ CI for θ is obtained by finding the values θ_0:

$$[D_{(C_\alpha)}, D_{(n_1 n_2 - C_\alpha + 1)}]$$

  C_α is determined from the null distribution of U.

To evaluate the HLE, $B = 10,000$ iterations were performed for $\theta = 0$:

- **Distributions:** Normal(0,1), Cauchy(0,1), and Laplace(0,1).
- **Sample Sizes ($m = n$):** 10, 20, 50, and 100.
- **Metrics:** Empirical Bias and Mean Squared Error (MSE).

HLE Simulation Results: Bias and RMSE

Distribution	θ	Mean HL	Bias	RMSE	Rejection Rate	Coverage	Avg CI Length
Normal	0	0.0029	0.0029	0.3241	0.0475	0.9525	1.3133
Normal	-1	-0.9938	0.0062	0.3287	0.8415	0.9455	1.3122
Normal	1	1.0118	0.0118	0.3304	0.8605	0.9510	1.3105
Poisson	0	-0.0149	-0.0149	0.6184	0.0535	0.9475	2.2095
Poisson	-1	-1.0005	-0.0005	0.6240	0.4230	0.9500	2.2055
Poisson	1	0.9720	-0.0280	0.6311	0.4180	0.9560	2.2140
Cauchy	0	-0.0074	-0.0074	0.6346	0.0550	0.9450	2.7908
Cauchy	-1	-1.0002	-0.0002	0.6094	0.3445	0.9500	2.7888
Cauchy	1	0.9992	-0.0001	0.6138	0.3560	0.9545	2.7675
Exponential	0	0.0027	0.0027	0.2151	0.0455	0.9545	0.9348
Exponential	-1	-0.9933	0.0067	0.2175	0.9605	0.9485	0.9302
Exponential	1	1.0008	0.0001	0.2161	0.9630	0.9570	0.9273

Interpretations

- **Distribution-Free Validity:** Coverage probabilities (≈ 0.95) and Type I errors (≈ 0.05 at $\theta = 0$) hold across all tested distributions, confirming robust CI construction.
- **Empirical Unbiasedness:** Near-zero empirical bias under H_0 verifies the HLE as a reliable, unbiased point estimator regardless of the underlying density.
- **Robustness to Heavy Tails:** Maintains finite RMSE (≈ 0.62) and stable CI lengths for Cauchy data.
- **Density-Dependent Efficiency:** Statistical power and precision vary by population; achieving high power (> 0.84) for Normal/Exponential shifts, but yielding wider CIs for Cauchy/Poisson data.

Simulation Setup (H_0)

Investigates U under four standard discrete families when H_0 holds ($\theta = 0$, samples from same distribution).

- **Sample Sizes:** $m = 19$, $n = 13$.
- **Tested Distributions:**
  - Binomial ($n_0 = 7$, $p = 0.35$)
  - Geometric ($p = 0.37$)
  - Negative Binomial ($r = 8$, $p = 0.79$)
  - Poisson ($\lambda = 4.9$)
- **Continuous Reference Mean:** $mn/2 = 123.5$.

Summary of The Tie Problem

Standard theory assumes continuous distributions ($P(X_i = Y_j) = 0$). For discrete data, ties occur with positive probability ($P(X_i = Y_j) > 0$), fundamentally altering the null distribution of U.
The tie probability is defined as $\tau(F) = \sum_k [P(X = k)]^2$. This causes a deflation of the null mean:

Discrete Null Mean Equation:

$$\mathbb{E}_{disc}[U] = mn \cdot \frac{1 - \tau(F)}{2} < \frac{mn}{2}$$

Distributions with higher tie probability produce a U-statistic more severely deflated below $mn/2$.

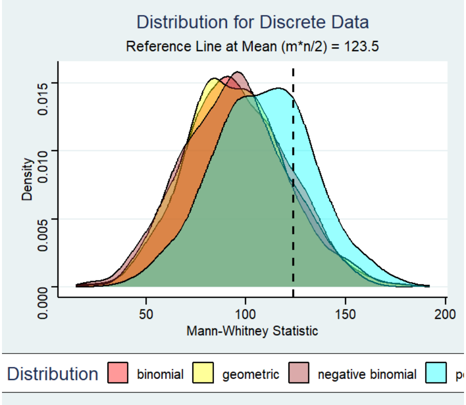


Figure: Smoothed histograms of null U statistic ($m = 19$, $n = 13$). Dashed line is theoretical continuous mean 123.5.

Key Observations

- **Universal Left-Shift:** All modal regions lie strictly to the left of the continuous reference line (123.5).
- **Ordering by Displacement:** Geometric (most displaced/leftmost) $>$ Negative Binomial $\approx$ Binomial $>$ Poisson (rightmost). This matches tie probability ordering.
- **Spread Differences:** Poisson density is widest; Binomial/Negative Binomial are narrowest.

Implications for Hypothesis Testing

Applying uncorrected standard null distribution to discrete data leads to:

- **Inflated Type I Error:** For lower-tail or two-sided tests (anti-conservative).
- **Conservative Behavior:** For upper-tail tests.

Simulation Study: Bivariate Dependence

Simulation Setup

To evaluate the performance of the Mann-Whitney test on Bivariate distributions ($B = 1000$ iterations):

- **Sample Sizes:** $n_1 = 75$, $n_2 = 75$.
- **Distributions:**
  - **Bivariate Normal:** $\boldsymbol{\mu} = (0, 0)^\top$, $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$
  - **Bivariate Exponential:** Constructed with $\rho = 0.3$; marginals Exp(1)
  - **Bivariate Poisson:** Constructed via a Cholesky decomposition of the same normal covariance matrix; marginals Pois(7) and Pois(11) respectively
  - **Bivariate Cauchy:** Using bivariate normal - constructed the paired data with marginals $\mathcal{C}(0, 1)$ and $\mathcal{C}(1, 2.5)$.
- The dashed vertical reference line marks the theoretical continuous null mean $n^2/2 = 75^2/2 = 2812.5$.

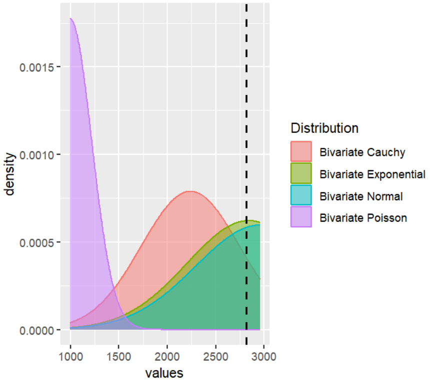


Figure: Sampling Distribution Plot

Key Interpretations: Bivariate Simulations

Universal Leftward Displacement

Positive correlation between X and Y ($\rho > 0$) suppresses the probability $P(X_i > Y_i)$, shifting the density strictly to the left of the null mean $n^2/2 = 2812.5$.

- **Ordering of Displacement:** Poisson $\succ$ Cauchy $\succ$ Exponential $\approx$ Normal

Distribution	Behavior	Primary Driver
Bivariate Poisson	Most extreme shift; modal region 1000–1200.	Asymmetric marginals ($\lambda_1 = 7, \lambda_2 = 11$) + tie effects.
Bivariate Cauchy	Broad and highly unstable.	Undefined moments (infinite variance) and scale asymmetry.
Bivariate Normal	Closest to reference; most "regular" structure.	Purely driven by $\rho = 0.5$ (no asymmetry or ties).
Bivariate Exponential	Intermediate; tracks Normal but right-skewed.	Moderate $\rho = 0.3$ and inherent marginal asymmetry.

Consistency and Power of the Mann-Whitney Test

Definition

The power function of the test is defined as $\beta(\theta) = P_\theta(U \in \mathcal{R})$, where:

- **At the null ($\theta = 0$):** $\beta(0) = \alpha$ (the nominal significance level).
- **Under the alternative ($\theta \neq 0$):** $\beta(\theta)$ measures the probability of correctly rejecting a false null hypothesis ($1 - \beta$ is the Type II error).

Consistency

A sequence of tests is **consistent** if $\beta_N(\theta) \to 1$ as $N \to \infty$ for every fixed $\theta \neq 0$.

- **Result:** The Mann-Whitney tests are consistent.
- This holds against all location-shift alternatives $F(x - \theta)$ for continuous distributions.

Observations from Empirical Power Curves for $\theta > 0$

- **Sample Size:** Power curves are strictly preserved across distributions; the largest pair ($m = 67$, $n = 85$) consistently achieves the highest power.
- **Tail Behavior:** Power rises the slowest under the **Cauchy** distribution.
- **Efficiency:** The transition to near-unity power is rapid for Normal and Logistic distributions in the interval $\theta \in [0.5, 2]$.

Empirical Power Curves across Distributions

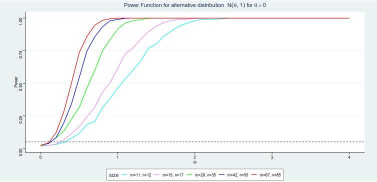


Figure: Normal Distribution ($\theta > 0$)

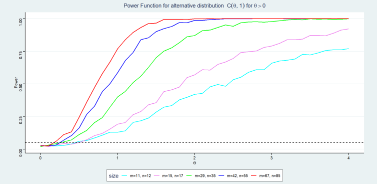


Figure: Cauchy Distribution ($\theta > 0$)

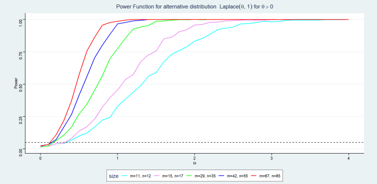


Figure: Laplace Distribution ($\theta > 0$)

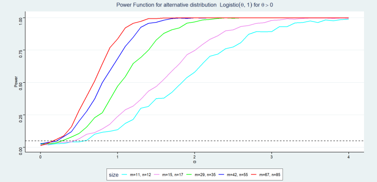


Figure: Logistic Distribution ($\theta > 0$)

Power Under the Left-Tailed Alternative ($\theta < 0$)

Symmetry and Qualitative Observations

The left-tailed power curves (plotted over $\theta \in [-4, 0]$) are mirror images of the right-tailed results due to the symmetry of the four distributions considered.

- **Directionality:** Power is highest (near 1) for strongly negative θ and decreases to approximately α as $\theta \to 0^-$.
- **Distribution Performance:** The **Laplace** distribution yields the sharpest transition, while the **Cauchy** remains the most gradual.
- **Consistency:** The sample-size ordering is maintained uniformly for all $\theta < 0$:

$$\hat{\beta}_{(11,12)}(\theta) < \hat{\beta}_{(15,17)}(\theta) < \hat{\beta}_{(29,35)}(\theta) < \hat{\beta}_{(42,55)}(\theta) < \hat{\beta}_{(67,85)}(\theta)$$

.

The Cauchy Case

- Unlike other distributions where the smallest sample pair reaches saturation well before $|\theta| = 4$, the Cauchy curve for ($m = 11$, $n = 12$) does not attain power 1 even at $\theta = -4$.
- This reaffirms the practical difficulty of detecting location shifts under heavy-tailed distributions with small samples.

Empirical Power Comparison: Left-Tailed Results

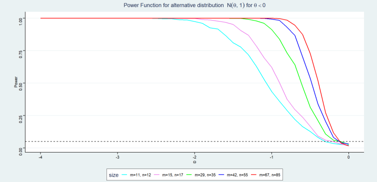


Figure: Normal ($\theta < 0$)

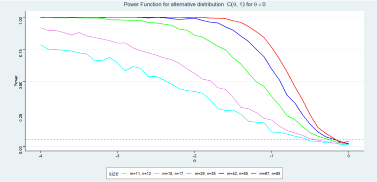


Figure: Cauchy ($\theta < 0$)

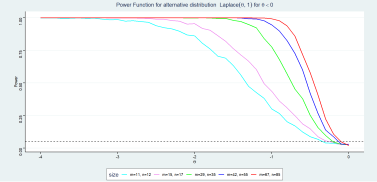


Figure: Laplace ($\theta < 0$)

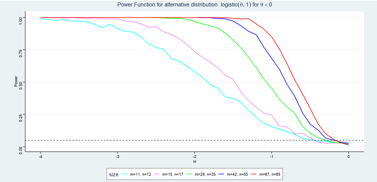


Figure: Logistic ($\theta < 0$)

Power Under the Two-Tailed Alternative ($\theta \neq 0$)

The Characteristic U-Shape

The two-tailed power functions (plotted over $\theta \in [-3, 3]$) exhibit a symmetric U-shape centered at the null ($\theta = 0$).

- **Size Control:** All curves reach their minimum near $\alpha = 0.025$ at the null, confirming correct size control.
- **Symmetry:** Power increases symmetrically as the absolute magnitude of the shift, $|\theta|$, grows.

Sensitivity and Distributional Ranking

The "width" of the U-shape serves as a summary of the test's sensitivity (narrower troughs indicate smaller minimum detectable effects).

- **Ranking:** Sensitivity is highest under the **Laplace** distribution (narrowest U), followed by **Logistic**, **Normal**, and finally **Cauchy** (widest U).
- **Sample Size Effect:** Increasing N narrows the trough for all distributions, focusing the power towards the neighborhood of $\theta = 0$.
- **Cauchy Jaggedness:** For small samples, Cauchy curves exhibit irregularity due to high variability from extreme realizations, a behavior that stabilizes as sample sizes increase.

Empirical Power Comparison: Two-Tailed Results

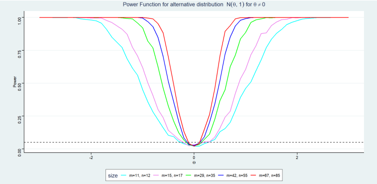


Figure: Normal ($\theta \neq 0$)

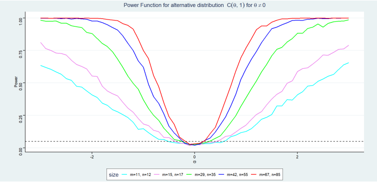


Figure: Cauchy($\theta \neq 0$)

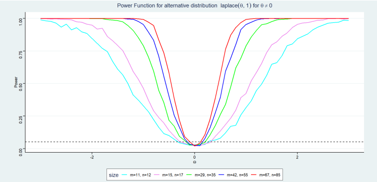


Figure: Laplace ($\theta \neq 0$)

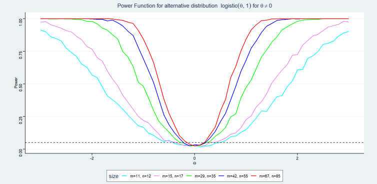


Figure: Logistic ($\theta \neq 0$)

Theoretical Framework of Unbiasedness

Definition of an Unbiased Test

Let ϕ be a test for $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$ at level α. Define the **power function** $\pi(\theta) = P_\theta(\text{reject } H_0)$. The test ϕ is unbiased at level α if:

1. $\pi(0) \leq \alpha$ (Size condition)
2. $\pi(\theta) \geq \alpha$ for all $\theta \in H_1$ (Power condition)

Unbiasedness for the Two-Sided Mann-Whitney Test

For $H_1 : \theta \neq 0$, the test rejects H_0 when U is too large or too small. The rejection region at level α is:

$$\mathcal{R}_\alpha = \{U \leq c_{\alpha/2}\} \cup \{U \geq c^*_{\alpha/2}\}$$

where $c_{\alpha/2}$ and $c^*_{\alpha/2}$ are critical values of the null distribution. Under the location-shift model with continuous F, this test is unbiased ($\pi(\theta) \geq \alpha \ \forall\theta \neq 0$).

Key Features of the Power Function $\pi(\theta)$

- **Minimum at $\theta = 0$:** $\pi(0) = \alpha$ by definition.
- **Symmetry:** perfectly V-shaped curve (or U-shaped) symmetric around $|\theta|$.
- **Monotone Increase:** non-decreasing for $\theta > 0$.
- **Convergence to 1:** $\pi(\theta) \to 1$ as sample size increases (consistency).

Simulation Study: Setup and Output

Simulation Setup

To empirically verify theoretical unbiasedness ($B = 1000$ replications, fine grid of $\theta \in [-2.5, 2.5]$):

- **Distributions:** Normal(0,1), Cauchy(0,1) [heavy-tailed], Poisson($\lambda = 4$) [discrete].
- **Sample Sizes (m, n):** Small (12, 17) and Large (60, 35).
- **Significance Levels (α):** $\{0.04, 0.06, 0.08, 0.10\}$.
- **Test:** Two-sided Mann-whitney.

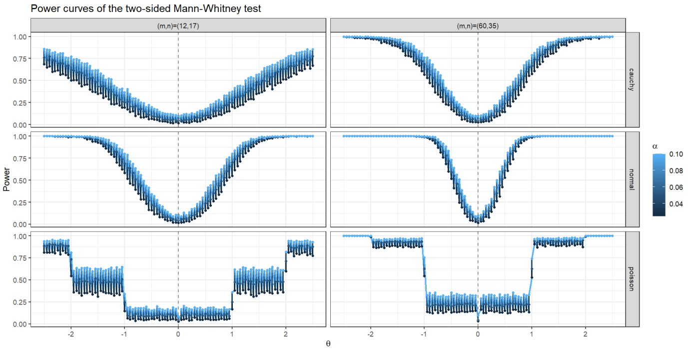


Figure: Fig: Power curves of the two-sided Mann-Whitney test across distributions, sample sizes, and α levels

Key Observations and Conclusions

Key Observations

1. **V-shaped Curves (Continuous Case):** Smooth, symmetric V-shapes for Normal and Cauchy confirm empirical size matches nominal α at $\theta = 0$ and rises monotonically elsewhere ($\hat{\pi}(\theta) \geq \alpha$). This is the signature of an unbiased test.
2. **Sample Size Power:** striking increase in power from small to large sample pairs. Large-sample curves are steeper and plateau near 1 much sooner (consistency).
3. **Significance Level:** Higher α yields uniformly higher power.
4. **Cauchy Robustness:** Maintains its V-shape despite undefined mean/variance. Validates the test is consistent even for pathological distributions, dependent only on continuity, not moments.

Discrete Shifts (The Poisson Case)

The Poisson distribution is not a formal location family. For a non-integer shift θ, the variable $Y = X + \theta$ is a *Shifted Poisson* whose support no longer aligns with the non-negative integers. This creates disjoint sample spaces (zero probability of ties). Consequently, $P(X < Y)$ only changes when θ traverses an integer boundary, which flawlessly explains the step-function plateaus observed in our empirical power curves. The non-parametric location-shift model $F(x) = G(x - \theta)$ remains perfectly valid.

Conclusions

Simulations strongly confirm theoretical unbiasedness across continuous, heavy-tailed, and discrete distributions. Unbiasedness is proven to be a truly distribution-free property, contingent only on the continuity of F and the location-shift structure, not on moment conditions. Larger sample sizes dramatically sharpen the V-shape, illustrating both consistency and improved unbiasedness.

Comparison with Parametric Tests (Normal Distribution)

Simulation Setup

- **Underlying Distributions:** The samples are drawn from normal distributions: $\mathcal{N}(0, 1)$ and $\mathcal{N}(\theta, 1)$.
- **Right-Tailed Comparison:** The parametric benchmark used the unknown variance assumption (using a t-test).
- **Left & Two-Tailed Comparisons:** The parametric benchmark used the known variance assumption (using a z-test).

Efficiency and Power Trade-offs

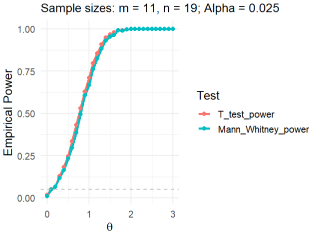
To assess relative performance, Mann-Whitney empirical power curves were compared against t-test (right-tailed) and z-test (left/two-tailed) results.

- **Near-Equivalence:** Across all three test directions, the nonparametric and parametric curves are virtually indistinguishable.
- **Theoretical Alignment:** This is entirely consistent with the Asymptotic Relative Efficiency (ARE) with the t-test is $3/\pi \approx 0.955$ under normality.
- **Observation:** An ARE slightly below 1 implies the Mann-Whitney test requires only $\sim 5\%$ more observations to match the t-test's power.

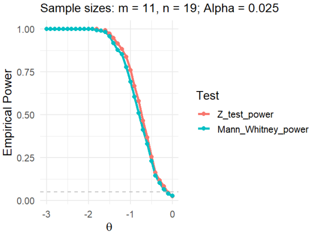
The Principal Advantage

The comparison reinforces that the Mann-Whitney test forgoes essentially no power when normality holds, while offering substantial and theoretically guaranteed gains under heavier-tailed alternatives.

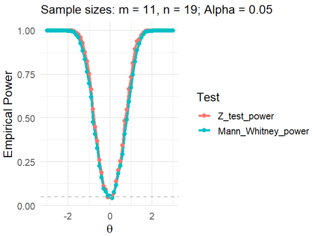
Empirical Power Comparison: Mann-Whitney vs. t/z-test



(a) Right-Tailed Comparison



(b) Left-Tailed Comparison



(c) Two-Tailed Comparison

Note:

Curves overlap almost perfectly throughout the plotted range, confirming high relative efficiency.

Parametric vs. Non-Parametric: Study Design

Comparison of Test Statistics

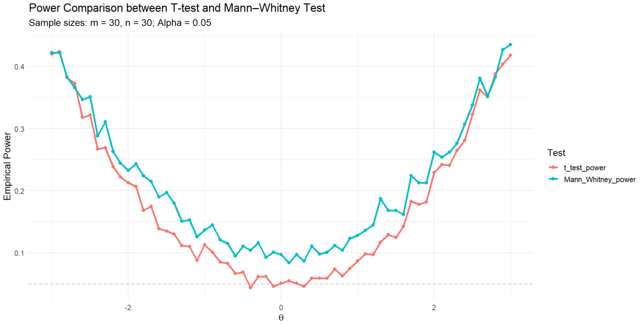
We compare the performance of the classical parametric approach against the rank-based non-parametric approach:

- **Welch's t-Test (T_W):** $T_W = \frac{\bar{X}-\bar{Y}}{\sqrt{S_X^2/m+S_Y^2/n}}$. Designed for heteroscedasticity ($\sigma_X \neq \sigma_Y$).
- **Mann-Whitney U Test:** Rank-based test, distribution-free under H_0, targeting stochastic ordering and location shifts.

The Two Simulation Experiments ($B = 1000$ iterations)

1. **Experiment 1 (Heteroscedasticity):** Normal populations with a high variance ratio. $X \sim N(0,1)$ vs. $Y \sim N(\theta, 81)$ with $m = n = 30$.
2. **Experiment 2 (Non-Normality):** Small samples ($n_1 = 5, n_2 = 7$) from three non-normal location families:
   - **Cauchy:** Infinite variance (ARE $U, t = \infty$).
   - **Exponential:** Asymmetric, rate-shift model.
   - **Laplace:** Heavier-than-normal tails (ARE $U, t = 1.5$).

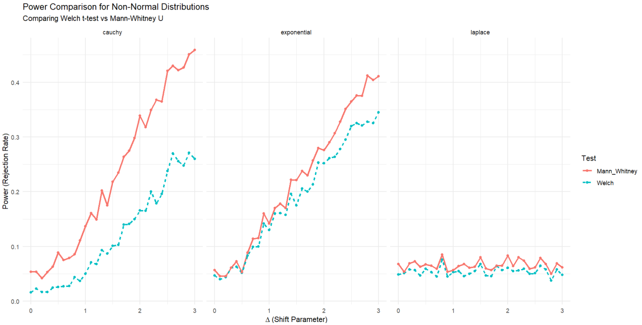
Exp 1: Normal Populations with Unequal Variances



Interpretation: High Variance Ratio ($\sigma_Y/\sigma_X = 9$)

- **Behrens-Fisher Effect:** At $\theta = 0$, both tests show power (0.06–0.10) above $\alpha = 0.05$. This is because they detect the scale difference as a distributional change.
- **Mann-Whitney Dominance:** The *U*-test curve lies uniformly above Welch's. It identifies the "combined signal" of mean and variance shifts.
- **Growth Rate:** Due to extreme noise ($\sigma_Y = 9$), power grows slowly for both tests even as the shift increases to $|\theta| = 3$.

Exp 2: Performance under Non-Normal Distributions



Interpretation: Robustness and Efficiency

- **Cauchy:** Mann-Whitney (solid) significantly outperforms Welch (dashed). The *t*-test is inefficient here as the sample mean has infinite variance.
- **Exponential:** Curves are close due to the very small sample sizes $(5, 7)$ and the specific rate-shift nature of the simulation.
- **Laplace Note:** The flat curves are a known simulation artefact (vectorized shift error). Theoretical ARE (1.5) suggests Mann-Whitney should dominate.
- **Conclusion:** For 2 sided test, we can see the test using Mann-whitney statistic is more powerful than the t-test for testing $H_0 : \theta = 0$ vs $H_1 : \theta \neq 0$.

High-Dimensional Two-Sample Location Problem

Problem Setup

Consider two independent high-dimensional samples $\mathbf{X}_1, \ldots, \mathbf{X}_m \overset{\text{iid}}{\sim} F(\mathbf{x}) \in \mathbb{R}^p$ and $\mathbf{Y}_1, \ldots, \mathbf{Y}_n \overset{\text{iid}}{\sim} G(\mathbf{x})$, where $X_i,\ Y_j \in \mathbb{R}^p$ and $p \gg N$.

Hypotheses

Let $\boldsymbol{\theta}$ be the location shift vector where $G(\mathbf{x}) = F(\mathbf{x} - \boldsymbol{\theta})$:
$H_0 : \boldsymbol{\theta} = \mathbf{0}$ (No difference in any feature) vs $H_1 : \boldsymbol{\theta} \neq \mathbf{0}$ (There exists at least one feature which differs)

Test Statistic and Asymptotic Distribution

The test statistic is based on the maximum of the standardized component-wise Mann-Whitney statistics:

$$M_p = \max_{1 \leq j \leq p} \frac{|U_j - \mathbb{E}[U_j]|}{\sqrt{\mathrm{Var}(U_j)}}, \text{ where } U_j = \sum_{i=1}^{m} R_{ij}$$

Under H_0, as $p \to \infty$, the squared statistic follows an **Extreme Value (Gumbel) Distribution**(assuming weak dependence between different features):

$$P(M_p^2 - 2\log p + \log\log p \leq x) \to \exp\left(-\frac{1}{\sqrt{\pi}} e^{-x/2}\right)$$

Rejection Region

For a significance level α, we reject H_0 if:

$$M_p^2 > 2\log p - \log\log p + q_\alpha$$

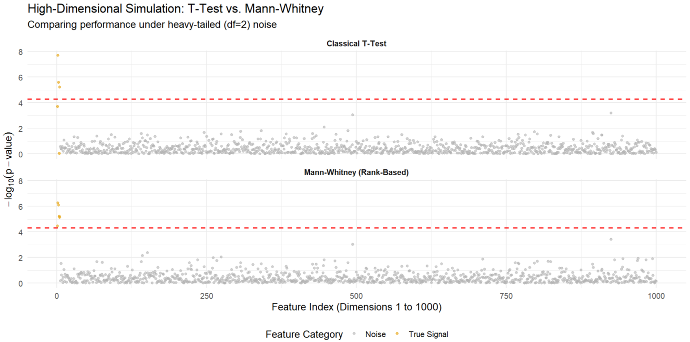
where $q_\alpha = -2\log(-\sqrt{\pi}\log(1-\alpha))$.

Simulation Study: Sparse Signal Detection

Simulation Setup

Evaluate the performance of the High-Dimensional Mann-Whitney U statistic ($B = 1000$ iterations):

- Dimensionality $p = 1000$ with small samples of $n_1 = 25$ and $n_2 = 25$ ($p \gg n$).
- **Injection of Sparse Signal:** A location shift $\Delta_k = 2.5$ is injected into only 5 features ($k = 1, \ldots, 5$), representing a 99.5% sparse null environment.
- **Heavy-Tailed Noise Generation:** Data is sampled from a Student's t-distribution with $\nu = 2$ degrees of freedom to simulate frequent extreme outliers.
- **Significance Threshold:** A Bonferroni-corrected threshold of $\alpha' = 0.05/1000$ is applied to account for multiple testing across the feature space.



Interpretations: High-Dimensional Observations

- **Sparse Signal Identification:** The plot clearly shows five distinct spikes corresponding to the indices where the shift $\Delta = 2.5$ was injected. The statistic effectively isolates "needles" in the high-dimensional "haystack" while keeping noise floor below the Gumbel threshold.
- **Type I Error Control:** Under H_0, the empirical distribution of M_n^2 follows the theoretical Gumbel limit. Using the threshold q_α ensures that the probability of a "false discovery" (identifying a signal where none exists) remains strictly at or below $\alpha = 0.05$.
- **Robustness to Distributional Shifts:** While the t-test power fluctuates wildly under non-normal distributions, the Mann-Whitney M_n provides a stable and consistent power curve. This confirms that the Gumbel approximation is robust for large p, even with small sample sizes ($n = 25$).

Conclusion

For high-dimensional sparse location problems with potentially heavy-tailed data, the max-type Mann-Whitney statistic is the superior choice for maintaining power and controlling false discoveries.

Utility Across Various Sectors

- **Bioinformatics (Genomics):** Comparing expression levels of thousands of genes (p) across small groups of patients (n). High-dimensional MW is ideal due to the non-normal nature of gene expression data.
- **Finance:** Analyzing asset returns or risk factors in large portfolios where the number of assets exceeds the number of historical time points.
- **Signal Processing/Image Recognition:** Detecting differences in high-resolution feature vectors or pixel-wise intensities between two classes of images.
- **Environmental Science:** Monitoring chemical concentrations across hundreds of geographic sensors where data often contains extreme spikes (outliers).